

using an underwater laser system, just by establishing line-of-sight.

Laser communication represents a mature, reliable approach for broadband access. Such systems have been engineered to provide robust performance that is highly competitive with other access approaches, offering high capacity, excellent availability of 99.9 per cent, lowest cost per bps, and rapid deployment in less than one hour. The use of an encoder and decoder allows the development of secure communication links.

Secure optical communication links are being extensively used in the defence industry to transmit secret information. These systems are compatible with a wide range of applications and markets, and are sufficiently flexible to be easily implemented using a variety of different architectures. Because of these features, market projections indicate healthy growth for optical wireless sales.

Microwaves cannot be used for underwater laser communication as they do not penetrate through water. Very-low-frequency sound waves transmit through water but they cannot enable high rates of data transfer. Blue-green lasers penetrate through water and are used for high-speed communication applications.

Inter-satellite links employ infrared laser beams for transmitting data from satellite to satellite. Lasers are used, as they do not suffer from attenuation in space and do not have precise pointing requirements.

The application potential of laser communication is expanding at a brisk pace and many new uses are emerging. Laser communication

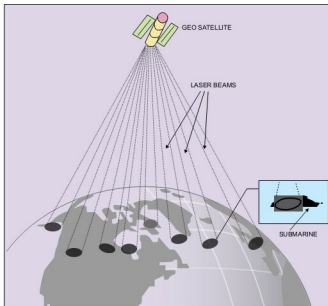


Fig. 4: Satellite-to-submarine communication

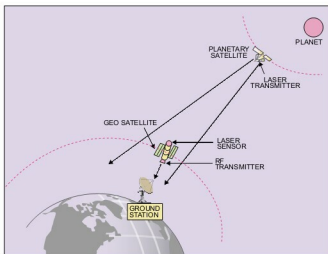


Fig. 5: An interplanetary TV link

is also being used for satellite-to-satellite links, satellite-to-submarine communication and interplanetary TV links.

Emerging trends in laser communication

Satellites can be used to communicate with many submarines that are submerged in sea water at depths of 100 metres or so. This would eliminate the need for submarines to come to the surface to establish communication, which would reduce their vulnerability. This concept is highlighted in Fig. 4. Satellites in geostationary

orbit transmit a large number of narrow blue-green laser beams to create random spots on the ocean, with each beam transmitting encrypted data. A large number of spots are generated, creating empty positions so as not to give away the location of the submarines. Blue-green laser is used for maximal penetration in sea water.

The concept of the interplanetary TV link is shown in Fig. 5. The setup makes use of a satellite orbiting around a planet, with which the link has to be established, and a satellite moving in a geostationary orbit around the earth. The planetary satellite makes use of a low-power laser to transmit signals. The earth-orbiting satellite will have a sensor to receive the optical signal, process it and convert it into microwave signals. The signal is converted from the optical spectrum to the microwave spectrum, as the optical signals do not penetrate clouds and are highly attenuated by rain. The conversion therefore allows the establishment of a non-interruptive link, which enables the monitoring of events happening on different planets on a real-time basis.

Underwater laser communication is an immensely interesting and attractive research topic for scholars and academics. Many advanced countries around the world are trying to successfully implement the technology for military purposes. Successful implementation of this technology for long-distance use is a challenging and uphill task; however, if achieved, it will be a major step forward in the field of communication. ●

The author is a technical editor trainee with Wiley Publications